RAG (Retrieval-Augmented Generation)

Short Questions & Answers • Interview / Viva Revision Notes

1. RAG Fundamentals

1. What is RAG?

RAG stands for Retrieval-Augmented Generation. It retrieves relevant information from an external knowledge source and gives it to an LLM to generate a more grounded answer.

2. Why is RAG used?

• To answer questions using private or domain-specific documents.
• To reduce unsupported or made-up answers.
• To update knowledge without retraining the model.

3. What are the main steps in RAG?

1. Load documents.
2. Split them into chunks.
3. Convert chunks into embeddings.
4. Store embeddings in a vector database.
5. Retrieve relevant chunks for a question.
6. Give the question and retrieved context to the LLM.

4. What is the difference between RAG and fine-tuning?

RAG provides relevant external context at query time. Fine-tuning changes model parameters through additional training. RAG is often better for frequently changing document knowledge; fine-tuning is useful for behavior, style, or task adaptation.

5. Does RAG train the LLM?

No. In a typical RAG system, the LLM is not retrained. Documents are indexed separately, and relevant content is supplied when a question is asked.

2. Documents, Chunking & Embeddings

6. What is document loading?

It is the process of reading source material such as PDFs, Word files, web pages, or text files so the system can process their content.

7. What is chunking?

Chunking divides long documents into smaller text sections. Smaller, meaningful chunks are easier to search and fit within the LLM's context window.

8. What is chunk size and overlap?

Chunk size is the amount of text in each chunk. Overlap is the repeated text shared between adjacent chunks. Overlap helps preserve context across chunk boundaries.

9. What are embeddings?

Embeddings are numerical vectors representing the meaning of text. Texts with similar meanings tend to have vectors that are close together in vector space.

10. What is a vector database?

A system that stores embeddings and helps find vectors similar to a query. Examples include FAISS, Chroma, Pinecone, and Weaviate.

11. What is semantic search?

Semantic search finds content based on meaning rather than only exact keyword matches. For example, a search for “reset my password” may retrieve a section titled “account recovery.”

3. Retrieval & Generation

12. What is a retriever?

A retriever searches the indexed knowledge base and returns the most relevant chunks for the user's question.

13. What is top-k retrieval?

It returns the k highest-ranked results. For example, top-k = 4 asks the retriever to return the four most relevant chunks.

14. What is similarity search?

It ranks text chunks by how close their embeddings are to the query embedding, using a measure such as cosine similarity or Euclidean distance.

15. What is a prompt in RAG?

The prompt combines instructions, the user's question, and retrieved context. It guides the LLM to answer using the supplied evidence.

16. What is hallucination, and how can RAG help?

A hallucination is an answer that sounds plausible but is false or unsupported. RAG can reduce this by supplying relevant source material, but it cannot eliminate hallucinations completely.

17. What happens if retrieval returns irrelevant chunks?

The final answer may be inaccurate. Improve chunking, embeddings, metadata filters, query rewriting, retrieval settings, or add reranking.

18. What is reranking?

Reranking takes an initial set of retrieved chunks and reorders them with a stronger relevance model before passing the best results to the LLM.

4. Tools, Architecture & Evaluation

19. What is LangChain?

LangChain is a framework for building LLM applications. It provides components for document loading, text splitting, embeddings, retrievers, prompts, and model calls.

20. What is FAISS?

FAISS is a library for efficient similarity search over dense vectors. It is commonly used to build a local vector index.

21. What is an LLM's role in RAG?

The LLM reads the retrieved context and the user's question, then produces a natural-language response. The retriever finds evidence; the LLM generates the answer.

22. What is a basic RAG pipeline?

PDF/Text files → text extraction → chunking → embeddings → vector store → retrieval → prompt with context → LLM response.

23. How do you evaluate a RAG system?

- Retrieval: Are the relevant chunks found?
- Faithfulness: Is the answer supported by retrieved context?
- Relevance: Does it answer the question?
- Latency and cost: Is it fast and affordable?

24. What are common RAG challenges?

Poor PDF extraction, bad chunk boundaries, irrelevant retrieval, outdated documents, large context, slow responses, and hallucinations.

25. How can you improve RAG performance?

Use clean text extraction, meaningful chunk sizes, suitable embeddings, metadata filters, hybrid search, reranking, clear prompts, source citations, and evaluation with representative questions.

26. What is hybrid search?

Hybrid search combines keyword-based retrieval with semantic vector search. It can work well when both exact terms and meaning matter.

27. Can RAG work with PDFs?

Yes. A PDF loader extracts text; the text is chunked, embedded, indexed, and retrieved when users ask questions. Scanned PDFs may require OCR.

28. What is conversational RAG?

Conversational RAG uses chat history to understand follow-up questions, retrieves relevant information, and answers in the context of the ongoing conversation.